

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - II Linked List Manipulation:

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Class : Sy-IT	Batch : 2
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Write a program to:

1. Create a singly linked list.

Code:

```
class Node:
```

```
    def __init__(self,data):
```

```
        self.data = data
```

```
        self.next = None
```

```
node1 = Node(10)
```

```
node2= Node(11)
```

```
node3 = Node(12)
```

```
node4 = Node(14)
```

```
node1.next = node2
```

```
node2.next= node3
```

```
node3.next= node4
```

```
def traver(head):
```

```
curr=head
while (curr):
    print(curr.data, end=="=>")
    curr = curr.next
print("Null")
```

```
traver(node1)
print("Mehek Sawant")
```

Output:

```
= RESTART: C:/Users/DELL/OneDrive,
DS/prac2.py
10=>11=>12=>14=>Null
Mehek Sawant
>>
```

2.(a) Insert a node at the beginning

Code:

```
class Node:
    def __init__(self,data):
        self.data = data
        self.next = None
```

```
node1 = Node(10)
node2= Node(11)
node3 = Node(12)
node4 = Node(14)
```

```
node1.next = node2
```

```
node2.next= node3
```

```
node3.nest= node4
```

```
global head
```

```
head = node1
```

```
def traver(head):
```

```
    curr=head
```

```
    while (curr):
```

```
        print(curr.data, end==">")
```

```
        curr = curr.next
```

```
    print("Null")
```

```
def insert_beg(data):
```

```
    global head
```

```
    new= Node(data)
```

```
    new.next = node1
```

```
    head = new
```

```
insert_beg(55)
```

```
traver(head)
```

Output:

```
>> |
    = RESTART: C:/Users/DELL/OneD
    DS/prac2.py
    55=>10=>11=>12=>Null
>> |
```

(b) end

Code:

```
class Node:
```

```
    def __init__(self,data):
```

```
        self.data = data
```

```
        self.next = None
```

```
node1 = Node(10)
```

```
node2= Node(11)
```

```
node3 = Node(12)
```

```
node4 = Node(14)
```

```
node1.next = node2
```

```
node2.next= node3
```

```
node3.nest= node4
```

```
global head
```

```
head = node1
```

```
def traver(head):
```

```
    curr=head
```

```
    while (curr):
```

```
    print(curr.data, end="=>")
    curr = curr.next
print("Null")
```

#insert at start

```
def insert_beg(data):
```

```
    global head
```

```
    new= Node(data)
```

```
    new.next = node1
```

```
    head = new
```

#insert at end

```
def insert_end(data):
```

```
    global head
```

```
    new = Node(data)
```

```
    if head is None:
```

```
        head = new
```

```
    else:
```

```
        temp= head
```

```
        while temp.next:
```

```
            temp= temp.next
```

```
        temp.next= new
```

```
insert_beg(55)
```

```
insert_end(34)
```

```
traver(head)
```

Output:

```

>>>
= RESTART: C:/Users/DELL/OneD
DS/prac2.py
55=>10=>11=>12=>34=>Null
...

```

(c) and at a given position in a linked list.

Code:

```
class Node:
```

```
    def __init__(self,data):
```

```
        self.data = data
```

```
        self.next = None
```

```
node1 = Node(10)
```

```
node2= Node(11)
```

```
node3 = Node(12)
```

```
node4 = Node(14)
```

```
node1.next = node2
```

```
node2.next= node3
```

```
node3.nest= node4
```

```
global head
```

```
head = node1
```

```
def traver(head):
```

```
    curr=head
```

```
    while (curr):
```

```
    print(curr.data, end="=>")
    curr = curr.next
print("Null")
```

#insert at start

```
def insert_beg(data):
```

```
    global head
```

```
    new= Node(data)
```

```
    new.next = node1
```

```
    head = new
```

#insert at end

```
def insert_end(data):
```

```
    global head
```

```
    new = Node(data)
```

```
    if head is None:
```

```
        head = new
```

```
    else:
```

```
        temp= head
```

```
        while temp.next:
```

```
            temp= temp.next
```

```
        temp.next= new
```

#insert at a certain position

```
def insert_position(data,pos):
```

```
    global head
```

```
    new = Node(data)
```

```
temp=head
for i in range (pos-2):
    temp = temp.next
new.next= temp.next
temp.next = new
```

```
insert_beg(55)
insert_end(34)
insert_position(21,2)
traver(head)
```

Output:

```
= RESTART: C:/Users/DELL/OneDrive/
DS/prac2.py
55=>21=>10=>11=>12=>34=>Null
```

3. Delete a node from a given position in a linked list.

Code:

```

#delete from the given position
def delete_pos(pos):
    global head

    temp=head
    temp=head
    for i in range (pos-2):
        temp = temp.next
    temp.next=temp.next.next

insert_beg(55)
insert_end(34)
insert_position(21,2)
delete_pos(4)
traver(head)
print("Mehek Sawant")

```

Output:

```

>
= RESTART: C:/Users/DELL/One
DS/prac2.py
55=>21=>10=>12=>34=>Null
Mehek Sawant

```

Curiosity Questions:

1. Why is inserting a node at the beginning of a singly linked list faster than inserting at the end?
2. What will happen if we don't update the head pointer when inserting a node at the beginning?
3. Why do we need to use a loop when inserting a node at a specific position in the linked list?
4. What would happen if you forgot to update the next pointer during deletion at a given position?
5. What should you check before deleting a node from a position in the linked list? How would you ensure that your program does not crash with invalid positions or empty list operations?